

FLOATING POINTS $F(\beta, t, L, U)$

SYSTEM

Where we say $x \in F(\beta, t, L, U) \iff x = \pm (\underbrace{d_1 \beta^s}_{\text{SIGN}} + \dots + \underbrace{d_t \beta^0}_{\text{MAINTISSA}}) \beta^p$ where $d_i \in [0, \beta-1]$
 $p \in [L, U]$

[NORMALIZED when $d_1 \neq 0$]

IEEE Single-Precision $\longrightarrow F(2, 24, -128, 127)$

IEEE Double-Precision $\longrightarrow F(2, 52, -1024, 1023)$

PROPERTIES

1) Finite and discrete

2) $TOT_NUM = 2(\beta-1)\beta^{t-1}(U-L+1) + 1$

3) $SMALLEST_NUM = \beta^{L-1}$

4) $BIGGEST_NUM = \beta^U(1-\beta^{-t})$

5) $MACHINE_PRECISION = \beta^{1-t}$

LINEAR ALGEBRA

SOLVING LINEAR SYSTEMS WITH DIRECT METHODS

1) **LOWER-UPPER FACTORIZATION**

$A = L \cdot U \longrightarrow LUx = b \longrightarrow \begin{cases} Ly = b & 1^\circ & O(n^2) \\ Ux = y & 2^\circ & O(n^2) \end{cases}$

LOWER TRIANGULAR $L_{ij} = 0 \quad j > i$

UPPER TRIANGULAR $U_{ij} = 0 \quad i > j$



This method can be very unstable because of the large rounding errors when matrices have small pivots close to 0 that are used in divisions for computing L and U.

PERMUTATION MATRIX

2) **GAUSSIAN FACTORIZATION**

$\uparrow$ $PA = L \cdot U \longrightarrow LUx = Pb \longrightarrow \begin{cases} Ly = Pb & 1^\circ \\ Ux = y & 2^\circ \end{cases}$

In this method, the permutation matrix is obtained by exchanging the rows on the identity matrix wrt the rows of A, so that by applying PA we obtain A with rows in a proper position.

In this way we overcome the previous error.

3) **CHOLESKY FACTORIZATION**

$A = L \cdot L^T \longrightarrow L \cdot L^T x = b \longrightarrow \begin{cases} Ly = b & 1^\circ \\ L^T x = y & 2^\circ \end{cases}$

LOWER TRIANGULAR UPPER TRIANGULAR

This method holds when A is symmetric and positive-definite, this can be seen as a special case of ①. Here we compute just one matrix L and its transpose.

SOLUTION OF A LINEAR SYSTEM

(not numerical method)

$$\begin{cases} a_{11}x_{11} + \dots + a_{1n}x_{1n} = b_1 \\ \vdots \\ a_{m1}x_{11} + \dots + a_{mn}x_{1n} = b_m \end{cases} \xrightarrow{m \times n} Ax = b \xrightarrow{m=n} \text{system has unique solution} \iff A \text{ is non singular}$$

$x = A^{-1}b \quad O(\cdot) \geq O(n^3)$

$\exists A^{-1} \iff \det(A) \neq 0$

ILL-CONDITIONING

When we solve a linear system $Ax=b$ we would look for A^{-1} and obtain $x=A^{-1}b$ but even if this is a stable algorithm this is not GUARANTEED to work properly, producing a not accurate solution. This happens when our matrix A is ill-conditioned, so the numbers in A are very small or very big and lead to perturbations with the floating point system. Those errors are emphasized by the operation in computing A^{-1} .

CONDITION NUMBER

$$\kappa_p(A) = \|A\|_p \cdot \|A^{-1}\|_p$$

CONDITIONING since $\kappa_p(A) \propto \dim(A)$ we can say A is:

- 1) ill-cond $\iff \kappa_p(A(n)) \approx c \cdot 10^n \quad c > 0$
- 2) well-cond $\iff \kappa_p(A(n)) \approx c \cdot n \quad c > 0$

ACCURACY AND ILL-CONDITIONING

$$\frac{\|\delta x\|_p}{\|x\|_p} \leq \kappa_p(A) \cdot \left(\frac{\|\delta b\|_p}{\|b\|_p} + \frac{\|\delta A\|_p}{\|A\|_p} \right)$$

PROOF:

0) We consider small perturbations on $Ax=b$

$$(A + \delta A)(x + \delta x) = b + \delta b$$

1) Expand and simplify

$$Ax + A\delta x + x\delta A + \delta A\delta x = b + \delta b \quad Ax=b$$

2) Isolate δx and assume $\exists A^{-1}$

$$A\delta x = \delta b - A^{-1}\delta A x \longrightarrow \delta x = A^{-1}\delta b - A^{-1}\delta A x$$

3) Apply p-norm and consider properties of norms: Δ inequality + sub-multiplicative

$$\|\delta x\|_p = \|A^{-1}\delta b - A^{-1}\delta A x\|_p \leq \|A^{-1}\delta b\|_p + \|A^{-1}\delta A x\|_p \leq \|A^{-1}\|_p \cdot \|\delta b\|_p + \|A^{-1}\|_p \cdot \|\delta A\|_p \cdot \|x\|_p$$

4) Divide by $\|x\|_p$

$$\frac{\|\delta x\|_p}{\|x\|_p} \leq \frac{\|A^{-1}\|_p \cdot \|\delta b\|_p}{\|x\|_p} + \frac{\|A^{-1}\|_p \cdot \|\delta A\|_p \cdot \|x\|_p}{\|x\|_p} = \|A^{-1}\|_p \cdot \left(\frac{\|\delta b\|_p}{\|x\|_p} + \|\delta A\|_p \right)$$

5) Insert $\kappa_p(A) = \|A\|_p \cdot \|A^{-1}\|_p$ and consider $\|b\|_p = \|Ax\|_p \leq \|A\|_p \cdot \|x\|_p$

$$\frac{\|\delta x\|_p}{\|x\|_p} \leq \frac{\kappa_p(A)}{\|A\|_p} \cdot \left(\frac{\|\delta b\|_p}{\|x\|_p} + \|\delta A\|_p \right) = \kappa_p(A) \cdot \left(\frac{\|\delta b\|_p}{\|b\|_p} + \frac{\|\delta A\|_p}{\|A\|_p} \right)$$

$\frac{\|b\|_p}{\|x\|_p} \geq 1$

ANALYTIC GEOMETRY

VECTORS NORM

MANHATTAN NORM

$$\|x\|_1 = \sum_{i=1}^n |x_i|$$

EUCLIDEAN NORM

$$\|x\|_2 = \sqrt{\sum_{i=1}^n x_i^2} = \sqrt{x^T x}$$

DOT PRODUCT

INFINITE NORM

$$\|x\|_\infty = \max_{1 \leq i \leq n} \{|x_i|\}$$

MATRIX NORM

FROBENIUS NORM

$$\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n a_{ij}^2}$$

2-NORM

$$\|A\|_2 = \sqrt{\rho(A^T A)}$$

$\rho(M)$ = largest among the absolute values of the eigenvalues of M

POSITIVE DEFINITE

$$\forall x \in V \setminus \{0\} : x^T A x > 0$$

- 1) A symmetric $\wedge (\lambda_i > 0 \forall \lambda_i \text{ in eigenvalues}(A)) \longrightarrow A$ is positive definite
 A symmetric $\wedge (\lambda_i \geq 0 \forall \lambda_i \text{ in eigenvalues}(A)) \longrightarrow A$ is positive semidefinite
- 2) $B^T B$ is always symmetric and positive semidefinite $\forall B \in \mathbb{R}^{m \times n}$
- 3) If $B \in \mathbb{R}^{m \times n}$ $\text{rank}(B) = \min(m, n) \longrightarrow A = B^T B$ is positive definite

ORTHOGONAL VECTORS



$$V \text{ and } W \text{ are ORTHOGONAL} \iff \langle V, W \rangle = 0 \iff \cos \theta = \frac{\langle V, W \rangle}{\|V\|_2 \|W\|_2} = 0 \quad [\theta = \pi/2 + k\pi]$$

$$V \text{ and } W \text{ are ORTHONORMAL} \iff \langle V, W \rangle = 0 \text{ and } \|V\|_2 = \|W\|_2 = 1$$

ORTHOGONAL MATRICES

$$A \in \mathbb{R}^{m \times n}, B \in \mathbb{R}^{m \times m} \text{ are ORTHOGONAL} \iff \text{ORTHONORMAL columns} \iff A^T A = I = A A^T \iff A^{-1} = A^T$$

⚠ LENGTH-CONSERVATIVE TRANSFORMATIONS

$$\|Ax\|_2 = \sqrt{(Ax)^T (Ax)} = \sqrt{x^T A^T A x} = \sqrt{x^T I x} = \sqrt{x^T x} = \|x\|_2$$

→ ROTATIONS

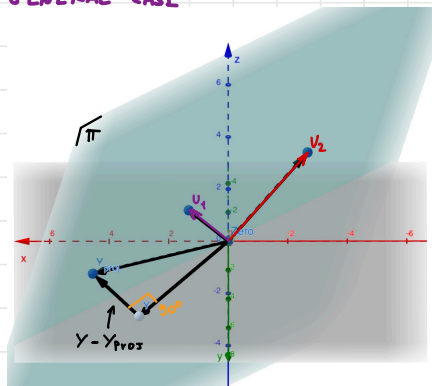
⚠ ANGLE-CONSERVATIVE TRANSFORMATIONS

$$\cos \theta = \frac{\langle Ax, Ay \rangle}{\|Ax\|_2 \|Ay\|_2} = \frac{(Ax)^T (Ay)}{\|Ax\|_2 \|Ay\|_2} = \frac{x^T A^T A^T y}{\|x\|_2 \|y\|_2} = \frac{x^T I y}{\|x\|_2 \|y\|_2} = \frac{x^T y}{\|x\|_2 \|y\|_2} = \frac{\langle x, y \rangle}{\|x\|_2 \|y\|_2}$$

ORTHOGONAL PROJECTION

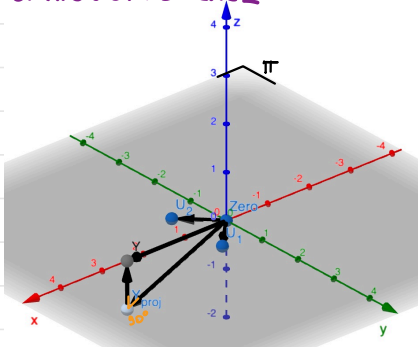
Given a matrix $A = \{a_1, \dots, a_n\}$ where $\{a_1, \dots, a_n\}$ are the basis vectors of the subspace U on which we want to project the vector $\vec{v}$, then $P_{\Pi}(v) = A \cdot (A^T A)^{-1} A^T = A \cdot A^T \iff A^T A = I \xrightarrow{A \in \mathbb{R}^{m \times n} \quad m \geq n}$
 $\xrightarrow{A^T A = I} A \in \mathbb{R}^{m \times n} \quad m > n$
 projection matrix $\xrightarrow{P_A}$

GENERAL CASE



$$\begin{aligned} U_1 &= [4/3, 8/3, 8/3]^T & U_2 &= [-8/3, -4/3, 8/3]^T \\ Y &= [3, 7, 1]^T & \Pi &= \text{span}(U_1, U_2) \\ A &= \begin{bmatrix} 4/3 & -8/3 \\ 8/3 & -4/3 \\ 8/3 & 8/3 \end{bmatrix} & \xrightarrow{P = AA^T} & P = \begin{bmatrix} 80/9 & 64/9 & -32/9 \\ 64/9 & 80/9 & 32/9 \\ -32/9 & 32/9 & 428/9 \end{bmatrix} \\ Y_{\text{proj}} &= P \cdot Y = \begin{bmatrix} 656/9 & 784/9 & 256/9 \end{bmatrix}^T \\ & & & \downarrow /16 \\ & & & \begin{bmatrix} 41/9 & 49/9 & 16/9 \end{bmatrix}^T \end{aligned}$$

ORTHOGONAL CASE



ORTHOGONAL BASIS $\xrightarrow{A \text{ is ORTHOGONAL}}$
 $\langle U_1, U_2 \rangle = 0$ and $\|U_1\| = \|U_2\| = 1$

$$\begin{aligned} U_1 &= [1/\sqrt{2}, 1/\sqrt{2}, 0]^T & U_2 &= [1/\sqrt{2}, -1/\sqrt{2}, 0]^T \\ Y &= [4, 3, 1]^T & \Pi &= \text{span}(U_1, U_2) \\ A &= \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \\ 0 & 0 \end{bmatrix} & \xrightarrow{P = AA^T = I} & P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ Y_{\text{proj}} &= P \cdot Y = [4, 3, 0]^T \end{aligned}$$

The orthogonality of $x - P_{\Pi}(x)$ wrt x ensures that $P_{\Pi}(x)$ is the nearest point, so it solves the problem
 $\min_{\Pi} \{ \|x - \Pi(x)\|_2^2 \} = P_{\Pi}(x)$

MATRIX DECOMPOSITION

Determinant (det(A))

- Multiplication:**
 $\det(\mathbf{AB}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$.
- Transpose:**
 $\det(\mathbf{A}^T) = \det(\mathbf{A})$.
- Identity matrix:**
 $\det(\mathbf{I}) = 1$.
- Singularity:**
 $\det(\mathbf{A}) = 0 \iff \mathbf{A}$ is not invertible.
- Scaling:**
 $\det(c\mathbf{A}) = c^n \det(\mathbf{A})$, where $n = \dim(\mathbf{A})$.
- Elementary operations:**
 - Row exchange: changes sign.
 - Row scaling by c : multiplies by c .
 - Row addition: no effect.
- Triangular/diagonal matrix:**
 $\det(\mathbf{A}) = \prod_{i=1}^n a_{ii}$.
- Inverse:**
 $\det(\mathbf{A}^{-1}) = \frac{1}{\det(\mathbf{A})}$, $\det(\mathbf{A}) \neq 0$.
- Eigenvalues:**
 $\det(\mathbf{A}) = \prod_{i=1}^n \lambda_i$, where λ_i are eigenvalues.

Trace (tr(A))

- Linearity:**
 $\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr}(\mathbf{A}) + \text{tr}(\mathbf{B})$,
 $\text{tr}(c\mathbf{A}) = c \text{tr}(\mathbf{A})$.
- Transpose:**
 $\text{tr}(\mathbf{A}^T) = \text{tr}(\mathbf{A})$.
- Identity matrix:**
 $\text{tr}(\mathbf{I}) = n$, for $\mathbf{I} \in \mathbb{R}^{n \times n}$.
- Cyclic property:**
 $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$ (valid even if $\mathbf{A}, \mathbf{B}$ are non-square).
- Eigenvalues:**
 $\text{tr}(\mathbf{A}) = \sum_{i=1}^n \lambda_i$, where λ_i are eigenvalues.
- Similarity invariance:**
 $\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{P}^{-1}\mathbf{A}\mathbf{P})$ for invertible $\mathbf{P}$.
- Rank relation:**
 $\text{tr}(\mathbf{A})$ depends only on non-zero eigenvalues.

WHY CHOLESKY IS USEFUL TO COMPUTE DETERMINANT

$$\begin{aligned} \det(\mathbf{A}) &= \det(\mathbf{L}\mathbf{L}^T) = \det(\mathbf{L}) \cdot \det(\mathbf{L}^T) = \det(\mathbf{L})^2 = \\ &= \left(\prod_{i=1}^n l_{ii} \right)^2 = \prod_{i=1}^n l_{ii}^2 \end{aligned}$$

EIGENVALUES and EIGENVECTORS

EIGENVALUE EQUATION

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x} \quad \text{for } \mathbf{A} \in \mathbb{R}^{n \times n}, \lambda \in \mathbb{C}, \mathbf{x} \neq \emptyset$$

IMPORTANT PROPERTIES

$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0$$

$$\mathbf{A} \text{ symmetric and positive definite} \implies \lambda \in \mathbb{R}, \lambda > 0$$

EIGENDECOMPOSITION AND DIAGONALIZATION

DIAGONAL MATRIX

$$\mathbf{A} = \begin{bmatrix} a_{11} & 0 & \dots & 0 \\ 0 & a_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn} \end{bmatrix}_{n \times n}$$

PROPERTIES

$$\det(\mathbf{A}) = \prod_{i=1}^n a_{ii}, \quad \det(\mathbf{A}^{-1}) = \prod_{i=1}^n a_{ii}^{-1}, \quad \mathbf{A}^{-1} = \begin{bmatrix} a_{11}^{-1} & 0 & \dots & 0 \\ 0 & a_{22}^{-1} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn}^{-1} \end{bmatrix}_{n \times n}$$

SVD of A
III

if $\mathbf{A}$ is symmetric $\implies$ diagonalization of $\mathbf{A}$
and positive semidefinite

DIAGONALIZABLE

$$\mathbf{A} \in \mathbb{R}^{n \times n} \text{ diagonalizable} \iff \exists \mathbf{P} \in \mathbb{R}^{n \times n} : \mathbf{D} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P} \quad (\mathbf{A} \text{ is similar to a diagonal matrix } \mathbf{D})$$

$$\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$$

EIGENDECOMPOSITION

columns of $\mathbf{P}$ are eigenvectors of $\mathbf{A}$

SINGULAR VALUE DECOMPOSITION (SVD)

⚠️ APPLICABLE TO ALL MATRICES AND ALWAYS EXISTS

Given $A \in \mathbb{R}^{m \times n}$: $r(A) \in [0, \min(m, n)]$

Then the SVD of A is $A = U \Sigma V^T$ where $U \in \mathbb{R}^{m \times m}$

$\Sigma \in \mathbb{R}^{m \times n}$ so that $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r$ and $\sum_{i=1}^r \sigma_i > 0$ and $\sum_{i=r+1}^{\min(m,n)} \sigma_i = 0$ ($i \neq j$)

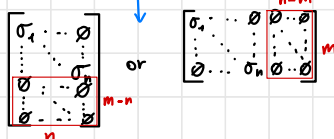
$V^T \in \mathbb{R}^{n \times n}$

$$\begin{matrix} m & & m & & n & & n \\ \boxed{A} & = & \boxed{U} & \boxed{\Sigma} & \boxed{V^T} \end{matrix}$$

LEFT-SINGULAR VALUES

SINGULAR VALUES

RIGHT-SINGULAR VALUES



Given $A \in \mathbb{R}^{m \times n}$ we can always build $A^T A \in \mathbb{R}^{n \times n}$ (symmetric and positive semidefinite) and diagonalize it.

$$A^T A = P \Lambda P^T = P \begin{bmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{bmatrix} P^T$$

if $P=V$ then $\lambda_i = \sigma_i^2$

n EIGENVALUES

$$A^T A = (U \Sigma V^T)^T (U \Sigma V^T) = V \Sigma^T U^T U \Sigma V^T = V \Sigma^T \Sigma V^T = V \begin{bmatrix} \sigma_1^2 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_n^2 \end{bmatrix} V^T$$

orthonormal eigenvectors are columns of V

⚡

$$A A^T = (U \Sigma V^T) (U \Sigma V^T)^T = U \Sigma V^T V \Sigma^T U^T = U \Sigma \Sigma^T U^T = U \begin{bmatrix} \sigma_1^2 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_n^2 \end{bmatrix} U^T$$

m EIGENVALUES

orthonormal eigenvectors are columns of U

GEOMETRICAL INTERPRETATION

Given an input vector x and an output vector y ($y = Ax$), then the SVD decomposition

$y = (U \Sigma (V^T x))^T$ can be seen as $x \xrightarrow{1} \text{rotated in } \mathbb{R}^n \xrightarrow{2} \text{scaled by } \sigma_i \xrightarrow{3} \text{rotated in } \mathbb{R}^m \rightarrow y$

We can write the SVD for each component: $A v_i = \sigma_i u_i$ $\forall i \in [1, r]$

$$A = \sum_{i=1}^r \sigma_i u_i v_i^T \quad \downarrow \text{from } r+1 \text{ to } n \text{ they are all } 0$$

We can now observe that:

$$\|A\|_2 = \sqrt{\lambda_{\max}(A^T A)} = \sqrt{\lambda_1} = \sqrt{\sigma_1^2} = \sigma_1$$

$$\|A^{-1}\|_2 = \sqrt{\lambda_{\max}((A^T A)^{-1})} = \sqrt{\lambda_{\min}(A^T A)^{-1}} = 1/\sigma_n$$

⚠️ How much error is introduced by a k -rank approximation?

$$\text{Given } \hat{A}(k) = \sum_{i=1}^k \sigma_i u_i v_i^T \text{ and } A = \sum_{i=1}^r \sigma_i u_i v_i^T$$

$$\text{Then } \|A - \hat{A}(k)\| = \sigma_{k+1}$$

Dim:

$$A - \hat{A}(k) = \sum_{i=1}^r \sigma_i u_i v_i^T - \sum_{i=1}^k \sigma_i u_i v_i^T = \sum_{i=k+1}^r \sigma_i u_i v_i^T = \sum_{i=k+1}^r \sigma_i A_i$$

$$\|A - \hat{A}(k)\|_2 = \left\| \sum_{i=k+1}^r \sigma_i A_i \right\|_2 = \sigma_{k+1} \quad (\text{biggest eigenvalue among the remaining ones})$$

NEW MATRIX THAT COULD BE DECOMPOSED WITH SVD AND THE FIRST SINGULAR VALUE WOULD BE σ_{k+1}

WEIGHTED COMPONENTS OF A

COMPRESSION FACTOR

It is the percentage of pixels we saved in memorizing X_k instead of X .

$$C_k = 1 - \frac{k(m+n+1)}{m \cdot n}$$

PROOF:

$$X \in \mathbb{R}^{m \times n} \longrightarrow mn \text{ pixels}$$

$$X_k = U_k \Sigma_k V_k^T = (m \times k) \cdot (k \times k) \cdot (k \times n) \xrightarrow{\text{FULL MATRICES = FALSE}} (mk) \cdot (k) \cdot (kn) = mk + k + kn \text{ pixels stored}$$

$$C_k = 1 - \frac{\text{pixels stored in } X_k}{\text{pixels stored in } X} = 1 - \frac{mk + k + kn}{mn} = 1 - \frac{k(m+n+1)}{mn}$$

PCA

$$X \in \mathbb{R}^d$$

We want to reduce the dimensionality of our data maintaining most of the information. In order to do so we need a projector $P: \mathbb{R}^d \longrightarrow \mathbb{R}^k$ with $k \ll d$.

1) So we define $Z = PX$ where $Z \in \mathbb{R}^{k \times N}$, $X \in \mathbb{R}^{d \times N}$, $P \in \mathbb{R}^{k \times d}$

2) We decompose X with SVD, so $X = U \Sigma V^T$ and we select the k -rank approximation $X_k = U_k \Sigma_k V_k^T$ where $U_k \in \mathbb{R}^{d \times k}$, $\Sigma_k \in \mathbb{R}^{k \times k}$, $V_k^T \in \mathbb{R}^{k \times N}$

PRINCIPAL COMPONENTS OF X

FEATURE VECTORS

3) We consider U_k as projection matrix $P = U_k^T \longrightarrow Z = U_k^T X$

We choose U_k and not V_k because U_k represents the basis of the row space of X , so it represents the row which is the whole sample with all features while V^T would represent the column space of X , so the whole feature along all samples.

CHAIN RULE

$$D[f(g(x))] = (\nabla f)(g(x)) \odot (\Delta g)(x)$$

CONTINUOUS OPTIMIZATION

OPTIMALITY CONDITIONS

- 1) x^* local minimum, f differentiable around x^* $\longleftrightarrow \nabla f(x^*) = \emptyset$ (x^* stationary point)
- 2) x^* local minimum, f twice differentiable around x^* $\longleftrightarrow \nabla f(x^*) = \emptyset$ and $\nabla^2 f(x^*)$ is positive semidefinite
- 3) f twice differentiable around x^* , $\nabla f(x^*) = \emptyset$ and $\nabla^2 f(x^*)$ is positive definite $\longleftrightarrow x^*$ is strict local minimum
HESSIAN

STRICT LOCAL MIN Given $x^* : \nabla f(x^*) = \emptyset$ $f(x^*) < f(x) \quad x \in [x^* - \epsilon, x^* + \epsilon]$ (strict local minimum)
 LOCAL MIN $f(x^*) \leq f(x) \quad x \in [x^* - \epsilon, x^* + \epsilon]$ (local minimum)

DIFFERENTIABILITY

$\exists \frac{\partial f(x_0)}{\partial x_i}$ and is continuous $\longleftrightarrow f$ differentiable in x_0

CONVEX FUNCTIONS

CONVEX SET $C \subset \mathbb{R}^n$ is convex $\longleftrightarrow \forall x, y \in C$ and $\forall \theta \in [0, 1] \subset \mathbb{R}$, $\theta x + (1 - \theta)y \in C$
EACH POINT BETWEEN x and y

CONVEX FUNC $f: C \rightarrow \mathbb{R}$ is convex $\longleftrightarrow \forall x, y \in C$ and $\forall \theta \in [0, 1]$,
 $f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$
 $\nabla^2 f$ is symmetric and positive semidefinite



⚠ MSE is convex

$$\text{MSE}(x) = \frac{\|Ax - b\|_2^2}{N} \longrightarrow \nabla \text{MSE} = \frac{2}{N} A^T (Ax - b) = \frac{2}{N} (A^T A x - A^T b) = \frac{2}{N} A^T A x - \frac{2}{N} A^T b$$

$$\nabla^2 \text{MSE} = \frac{2}{N} A^T A \longrightarrow \text{symmetric and positive semidefinite}$$

ITERATIVE ALGORITHMS

They work for generic $f: \mathbb{R}^n \rightarrow \mathbb{R}$ for computing local minima of f

DESCENT METHODS

$x_{k+1} = x_k + d_k p_k$ where the descent direction is chosen so that $f(x_{k+1}) < f(x_k)$, and in order to guarantee that, it must be $\nabla f(x_k)^T p_k < 0$

1) GRADIENT DESCENT

• $x_{k+1} = x_k - d_k \nabla f(x_k)$ we chose $p_k = -\nabla f(x_k)$ so that the constraint is always guaranteed, since

$$\nabla f(x_k)^T p_k = -\nabla f(x_k)^T \nabla f(x_k) = -\|\nabla f(x_k)\|_2^2 \leq 0 \quad \forall k$$

• $d_k \xrightarrow{\forall k} \text{CONSTANT} \rightarrow$ does not guarantee convergence

• **BACKTRACKING** $\rightarrow$ changes at each iteration and it's guaranteed to be correct, given $0 < c_1 < c_2 < 1$

1) **SUFFICIENT DECREASE** $\rightarrow$ ensure d_k not too small, by seeing if actual decrease is at least as a certain linear estimate.

$$f(x_k - d_k \nabla f(x_k)) \leq f(x_k) - c_1 d_k \|\nabla f(x_k)\|_2^2 \longrightarrow f(x_k) - f(x_k - d_k \nabla f(x_k)) \geq c_1 d_k \|\nabla f(x_k)\|_2^2$$

2) **CURVATURE CONDITION** $\rightarrow$ ensure d_k not too big, by checking that the new gradient in the actual direction is under a certain value.

$$\nabla f(x_k)^T \nabla f(x_k - d_k \nabla f(x_k)) \leq c_2 \|\nabla f(x_k)\|_2^2$$

• STOPPING CRITERIA

1) $\frac{\|\nabla f(x_k)\|_2}{\|\nabla f(x_0)\|_2} \leq \text{TOL}_f \longrightarrow$ the gradient is enough small to be considered 0 $\longrightarrow \nabla f(x^*) \approx 0$ x^* stationary point
 $\uparrow$

2) $\|x_k - x_{k+1}\|_2 \leq \text{TOL}_x \longrightarrow$ we are approximately still and so stuck

3) Set a max iterations

2) STOCHASTIC GRADIENT DESCENT

Similar to GD, instead of considering the whole dataset at each iteration we divide it in batches which contains random elements of the dataset and we use those subsets for computing gradients.

So given a model $f_\theta(x)$ we look for the best θ by $\theta_{k+1} = \theta_k - d_k \nabla_\theta \ell(\theta, M)$

⚠ The data points are randomly chosen and put into their respective minibatches before all the iterations, so that the points are considered exactly once.

PROBABILITY

RANDOM VARIABLE

$$X: \Omega \rightarrow T$$

DISCRETE X discrete $\iff T$ is finite or countable

PMF) $f_X: T \rightarrow [0, 1]$ and $f_X(x) = P(X=x)$

MEAN) $\mu = E(X) = \sum_{x \in T} x f_X(x)$

VAR) $\sigma^2 = \sum_{x \in T} (x - \mu)^2 f_X(x)$

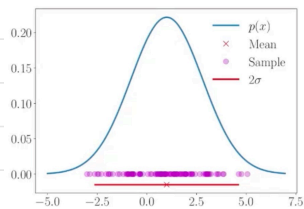
CONTINUOUS X continuous $\iff T = \mathbb{R}$

PMF) $f_X: T \rightarrow \mathbb{R}$ where $T = [a, b]$ and $P(X \in [a, b]) = \int_a^b f_X(x) dx$

MEAN) as discrete with integrals

VAR) as discrete with integrals

GAUSSIAN DISTRIBUTION



$$f_X(x) = \frac{e^{-\frac{(x-\mu)^2}{2\sigma^2}}}{\sigma\sqrt{2\pi}}$$

where $x \in \mathbb{R}$

BAYES THEOREM

$$P(\text{X} | \text{Y}) = \frac{\text{POSTERIOR}}{\text{EVIDENCE}} = \frac{\text{LIKELIHOOD} \cdot \text{PRIOR}}{P(\text{Y})}$$

VARIANCE

$$\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - E[X]^2$$

COVARIANCE

$$\text{Cov}(X, X) = \text{Var}(X)$$

$$\text{Cov}_{X,Y}[X, Y] = E_{X,Y}[(X - E_X[X])(Y - E_Y[Y])] = E[XY] - E[X]E[Y]$$

$$\overset{\text{matrix}}{\mathbb{R}^{D \times E}} \quad (\text{if } X \in \mathbb{R}^D \text{ and } Y \in \mathbb{R}^E) \quad \Rightarrow \quad \text{Cov}_X[X, X] \in \mathbb{R}^{D \times D} \text{ and is symmetric and positive semidefinite}$$

CORRELATION

$$\text{corr}[X, Y] = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}} \in [-1, 1]$$

MODELS AND DATA

ML PROCESS

DATA $x_n \in \mathbb{R}^D$ $n \in [1, N]$ in a dataset

MODEL $f: \mathbb{R}^D \rightarrow \mathbb{R}$ and $f(x, \theta) = \theta^T x + \theta_0$ where $\theta \in \mathbb{R}^D$

LEARNING Obtain the best θ so that the prediction $y = f(x_{\text{new}}, \theta)$ is a good prediction wrt some metric

MAXIMUM LIKELIHOOD ESTIMATION (MLE)

Given a dataset (X, Y) where $X \in \mathbb{R}^{N \times D}$, $Y \in \mathbb{R}^N$. We assume the samples x_i independent and identically distributed with probability $p(x)$ and the sampling will be $f(x_i) = y_i$ where $y_i \sim P(y|x_i)$.

Since $f_\theta(x) \approx f(x)$, we want to find the parameter θ that maximizes $P_\theta(y|x)$, this must be applied to the whole dataset $P_\theta(Y|X)$. We also make Gaussian assumption $P_\theta(y|x) \sim N(f_\theta(x), \sigma^2)$ and as a consequence, when we do the prediction $y = f_\theta(x) + \epsilon$ then $\epsilon \sim N(0, \sigma^2)$.

$$\begin{aligned} \theta_{MLE} &= \max_{\theta} \{P_\theta(Y|X)\} = \max_{\theta} \{P_\theta((y_1, \dots, y_N)|X)\} \stackrel{\text{indep.}}{=} \max_{\theta} \left\{ \prod_{i=1}^N P_\theta(y_i|x_i) \right\} \stackrel{\text{i.i.d.}}{=} \max_{\theta} \left\{ \prod_{i=1}^N P_\theta(y_i|x_i) \right\} = \\ &= \min_{\theta} \left\{ -\log \left(\prod_{i=1}^N P_\theta(y_i|x_i) \right) \right\} = \min_{\theta} \left\{ -\sum_{i=1}^N \log(P_\theta(y_i|x_i)) \right\} \stackrel{\text{Gaussian}}{=} \min_{\theta} \left\{ -\sum_{i=1}^N \log \left(\frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{-\frac{(y_i - f_\theta(x_i))^2}{2\sigma^2}} \right) \right\} = \\ &= \min_{\theta} \left\{ -\sum_{i=1}^N \left(\log \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right) - \frac{(y_i - f_\theta(x_i))^2}{2\sigma^2} \right) \right\} \stackrel{\text{const.}}{=} \min_{\theta} \left\{ \sum_{i=1}^N (y_i - f_\theta(x_i))^2 \right\} = \min_{\theta} \{ \|Y - f_\theta(X)\|_2^2 \} \end{aligned}$$

MAXIMUM A POSTERIORI ESTIMATION (MAP)

Now we try to maximize the probability of correct parameters θ , given the dataset X, Y , so $P(\theta|X, Y)$.

We will assume $P(y_i|x_i, \theta) \sim N(f_\theta(x_i), \sigma^2)$ and $P(\theta) \sim N(0, \sigma^2)$.

$$\begin{aligned} \theta_{MAP} &= \max_{\theta} \{P(\theta|X, Y)\} = \min_{\theta} \{-\log(P(\theta|X, Y))\} \stackrel{\text{BAYES}}{=} \min_{\theta} \left\{ -\log \left(\frac{P(X, Y|\theta) \cdot P(\theta)}{P(X, Y)} \right) \right\} \stackrel{\text{GENERAL PRODUCT RULE}}{=} \min_{\theta} \left\{ -\log(P(X, Y|\theta)) - \log(P(\theta)) \right\} = \\ &= \min_{\theta} \left\{ -\log \left(\frac{P(Y|X, \theta) \cdot P(X)}{P(X)} \cdot P(\theta) \right) \right\} \stackrel{\text{const.}}{=} \min_{\theta} \left\{ -\log(P(Y|X, \theta)) - \log(P(\theta)) \right\} = \\ &= \min_{\theta} \left\{ -\log \left(\prod_{i=1}^N P(y_i|x_i, \theta) \right) - \log(P(\theta)) \right\} = \min_{\theta} \left\{ -\sum_{i=1}^N \log(P(y_i|x_i, \theta)) - \log(P(\theta)) \right\} \stackrel{\text{Gaussian}}{=} \\ &= \min_{\theta} \left\{ \sum_{i=1}^N \frac{(y_i - f_\theta(x_i))^2}{2\sigma^2} - \log(P(\theta)) \right\} \stackrel{\text{const.}}{=} \min_{\theta} \left\{ \sum_{i=1}^N (y_i - f_\theta(x_i))^2 + \lambda \|\theta\|_2^2 \right\} \end{aligned}$$

LINEAR REGRESSION

We have a set of data $x_i \in \mathbb{R}^D$ and their corresponding noisy observations $y_i = f(x_i) + \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, \sigma^2)$, we want to infer f . Since the regression is linear, we use the linear equation $f(x) = x^T \theta$. Moreover, since it has to be linear in the parameters, we can consider non-linear transformations in the inputs $\Phi(x) \in \mathbb{R}^{M \times K}$ where $K-1$ is the degree and consider $f(x) = \Phi(x)^T \theta$.